Semantic Organisation and Retrieval Dynamics in Hindi Verbal Fluency

Mid-Project Analysis Report — Verbal Fluency Task (VFT)

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Abstract

This mid-project report presents experimental details, research hypotheses, planned analyses, and exploratory results from an ongoing study of Hindi verbal fluency at IIIT Hyderabad. The study uses a Verbal Fluency Task (VFT) across four semantic domains (Animals, Foods, Colours, Body-parts) administered to 35 participants (32~male, 3~female; $M_{\text{age}} = 23.1$~yrs, SD~= 1.9, range 19–27; 14 Indian states represented). A total of 712 valid Hindi responses were recorded. Responses were collected via a custom web experiment recorded in `responses.json`. Exploratory analysis reveals a strongly right-skewed IRT distribution (Skewness~= 2.54, Kurtosis~= 9.89), with mean 6490~ms and median 5389~ms. Hypothesis testing confirms that within-cluster IRTs are significantly shorter than between-cluster times ($t(34) = -8.91$, $p < .001$, $d = 1.51$), supporting the clustering-and-switching model [1]. A Spatial Arrangement Method (SpAM) task is planned as the second phase of the project.

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1 Introduction

1.1 Background and Motivation

The Verbal Fluency Task (VFT) is a cognitive paradigm in which participants freely recall category members within a fixed time window [1]. Behavioural signatures such as clustering (bursts of semantically related words) and switching (long pauses at subcategory boundaries) are well-documented in English-speaking populations but remain underexplored in Hindi. The present study applies the full BRSM statistical pipeline to characterise retrieval dynamics in a Hindi-speaking student cohort at IIIT Hyderabad. A Spatial Arrangement Method (SpAM) task [3] is planned as Phase 2.

1.2 Research Questions

Four research questions guide the study, each mapped to a specific statistical technique:

#	Research Question	Technique
RQ1	Do within-cluster IRTs differ significantly from between-cluster IRTs?	Welch's t-test, Cohen's d
RQ2	Does mean cluster size predict individual fluency scores?	Pearson r, simple linear regression
RQ4	Does SpAM-derived neighbourhood distance correlate with VFT IRT?	Pearson r, scatter plots *(planned)*

Note: A one-way ANOVA testing domain differences in IRT ($F(3, 708) = 2.18$, $p = .092$, $\eta^2 = .009$) was conducted exploratorily but yielded a non-significant, negligible effect and has been excluded from the main analysis pipeline.

2 Research Design

2.1 Experimental Design

The study employs a **within-subjects** repeated-measures design. The independent variable is semantic domain (nominal; 4 levels); the primary dependent variable is inter-response time (IRT, ratio scale, ms); the secondary DV is total words produced. Trial duration is fixed at 1~min per domain.

2.2 Participants and Demographics

Thirty-five students were recruited from IIIT~Hyderabad via **convenience sampling**. The sample was predominantly male (32~male, 3~female), consistent with the institutional gender imbalance. All participants were native or highly proficient Hindi speakers. A proportion of responses contained

code-mixed vocabulary, characteristic of this bilingual population, and was retained in the dataset. No participant reported a history of neurological or psychiatric disorder.

Demographic summary:

Variable	Value
N	35
Gender	32 Male, 3 Female
Age	$M = 23.1$~yrs, $SD = 1.9$, range 19–27
Education	$M = 16.5$~yrs, $SD = 1.7$, range 14–20
States (India)	14 states; Gujarat (7), MP (6), Bihar (5), Maharashtra (4+)

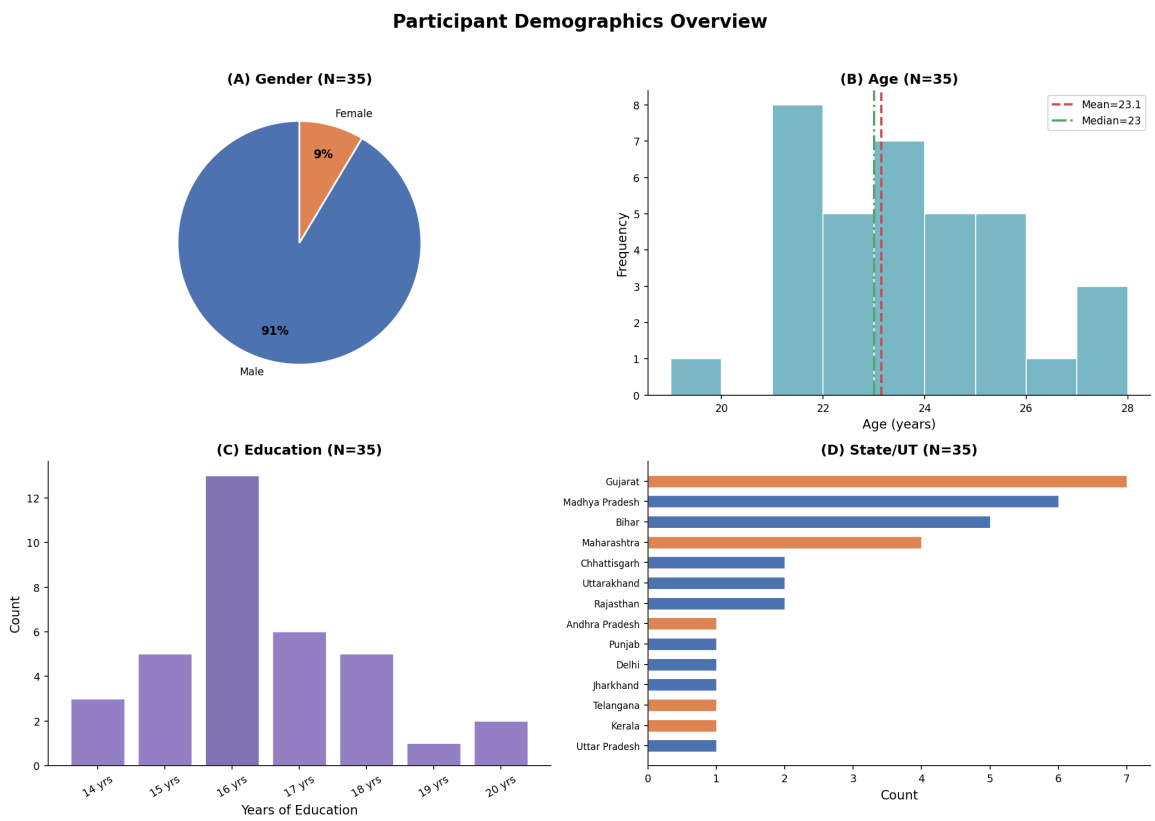

Figure 1: Figure 1. *Combined demographics panel.* Gender distribution (top-left), age histogram (top-right), years of education (bottom-left), geographic distribution across Indian states/UTs (bottom-right). Convenience sample from IIIT Hyderabad.

3 Materials, Procedure, and Data

3.1 Experimental Platform

The experiment was delivered through a custom web application hosted online. Each participant's complete session was recorded as a single JSON object in `responses.json`, keyed by a unique `session_id`. Each session is stored as a JSON object containing `subject_id`, `domain`, per-word `response_times` arrays (ms), `tagged_responses` (word + script tag), and `droppedwords` (SpAM spatial coordinates). IRTs were computed as $\text{IRT}_i = t_i - t_{i-1}$ per domain trial.

3.2 Verbal Fluency Task Procedure

Participants were presented with one category cue at a time (e.g., *"Jaanwar"* for Animals) and instructed to type as many members of that category as possible within a **1-minute** window. Each key-press (ENTER) was time-stamped to the nearest 100~ms. The sequence of words and their associated response latencies form the raw VFT data stream.

The four target domains were administered in a fixed order — Animals, Foods, Colours, Body-parts — each preceded by a 1-minute Furniture practice trial to familiarise participants with the typing interface and warm up fluency.

3.3 Data Processing

Raw JSON was parsed in Python to extract per-word IRTs ($\text{IRT}_i = t_i - t_{i-1}$). Devanagari responses were tagged `Hindi`; Latin-script entries `English`. The Hindi subset (53,% of tokens; $n = 712$) forms the analysis dataset. Clusters were identified following [1] using a per-participant adaptive switch threshold (mean IRT + 1~SD).

The merged file `merged_vft_spam_responses.csv` (1,040 rows) combines per-word IRT (`rt_ms`) with SpAM spatial coordinates (`x`, `y`), enabling the RQ4 cross-task correlation.

4 Exploratory Data Analysis

4.1 Response Counts and Data Coverage

Table~1 summarises Hindi response counts by domain.

Table 1: Table 1. Hindi response counts by domain.

Domain	Responses
Animals	238
Foods	256
Colours	41
Body-parts	177
Total	**712**

Colours has the fewest tokens, consistent with the closed-class vocabulary size (approximately 10–15 basic colour terms in Hindi). Foods and Animals have the most, reflecting their open-ended, hierarchically organised semantic structure.

5 Descriptive Statistics

5.1 Overall IRT Distribution

Table~2 presents descriptive statistics for all 712 valid Hindi IRTs. The distribution is strongly **right-skewed** (Skewness~= 2.54), with the mean (6490~ms) substantially exceeding the median (5389~ms). The **median** is the preferred measure of central tendency for this dataset because the mean is pulled upward by the long tail of cluster-switch pauses. High kurtosis (9.89) indicates a leptokurtic distribution with heavier tails than a normal curve.

Table 2: Table 2. Overall descriptive statistics for Hindi IRT ($n = 712$).

Statistic	Value (ms)
N	712
Mean	6489.5
Median	5389.4
Std Dev	5018.8
Min	732.8
Max	42634.4
IQR	4874.8
Skewness	2.54
Kurtosis	9.89

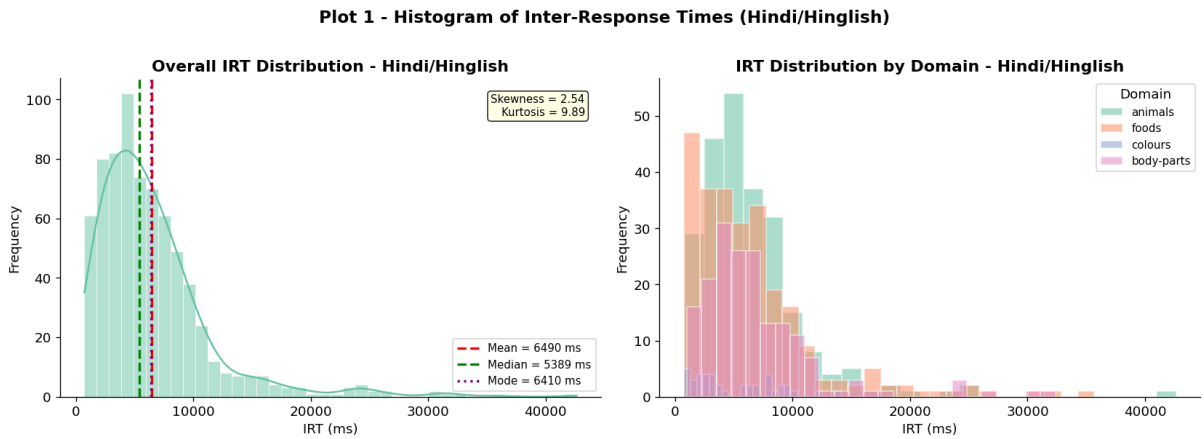

Figure 2: Figure 3. *Plot 1 — Histogram of Hindi IRT.* Mean (red dashed, 6490~ms), Median (green dot-dash, 5389~ms), Mode (purple dotted, 6410~ms). Mean > Median confirms positive skew. Domain-level histograms shown in right panel.

5.2 IRT by Semantic Domain

Table~3 decomposes the statistics by domain. Colours showed the lowest mean IRT (4975~ms) and near-zero skewness (0.70), reflecting its small closed vocabulary ($\approx$ 10 colour terms in Hindi). Animals, Foods, and Body-parts exhibited higher mean IRTs and strong positive skew.

Table 3: Table 3. IRT descriptive statistics by semantic domain.

Domain	N	Mean (ms)	Median (ms)	SD (ms)	Skew
Animals	238	6391	5414	4647	3.06
Body-parts	177	6872	5724	4994	2.51
Colours	41	4975	3484	3512	0.70
Foods	256	6559	5205	5525	2.28

6 Data Visualisation

Figure~4 shows the IRT distribution per domain via violin plots.

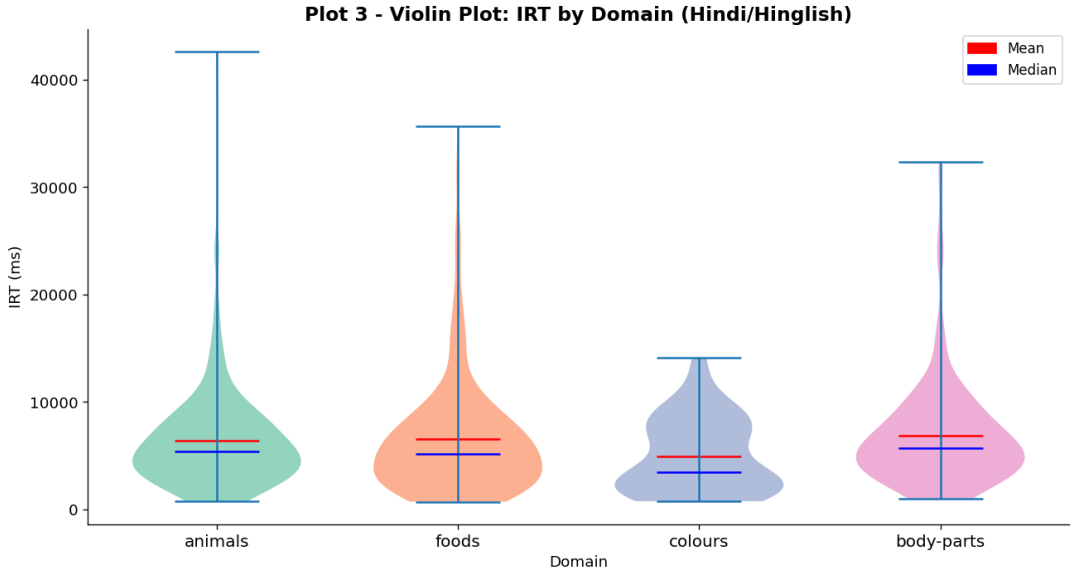

Figure 3: Figure 4. *Violin / box plots of IRT by domain.* Centre line = median; box = Q1–Q3; whiskers = 1.5 × IQR; dots = outliers. Colours is tightest; Body-parts is slowest.

Figure~5 plots IRT against word serial position within each domain. A positive slope per domain confirms the **lexical exhaustion effect**: as participants retrieve more words, the remaining items become less accessible and inter-response times lengthen.

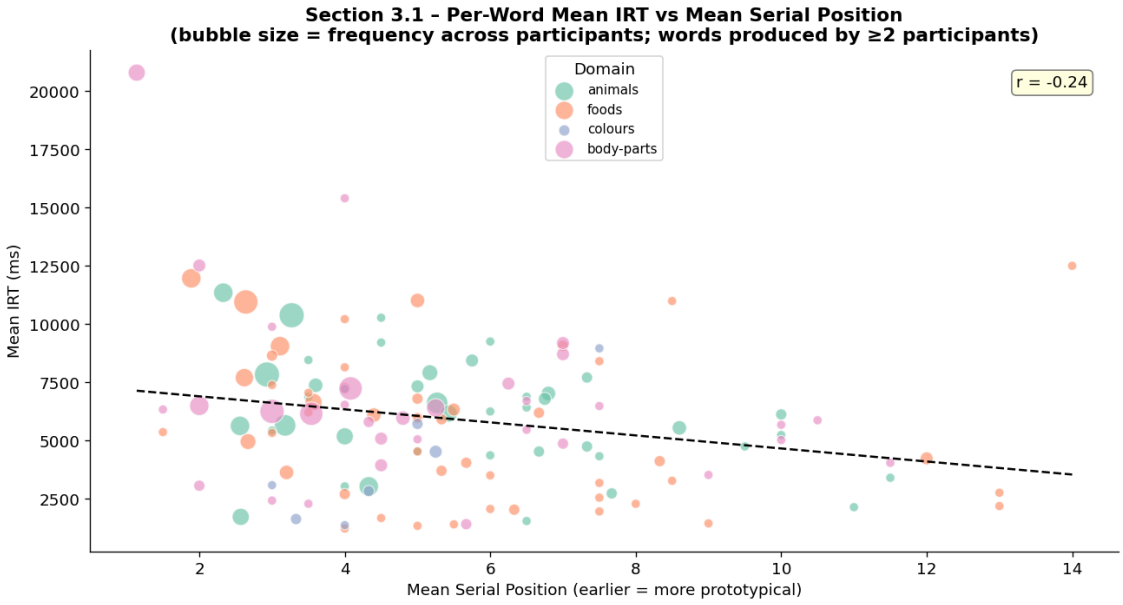

Figure 4: Figure 5. *Serial position vs IRT scatter plot.* Each point is one word produced by one participant. OLS trend line fit per domain. The positive slope in all four domains demonstrates that later-position words are retrieved more slowly — consistent with progressive depletion of the accessible semantic neighbourhood.

Figure~6 summarises cluster scoring across participants and domains.

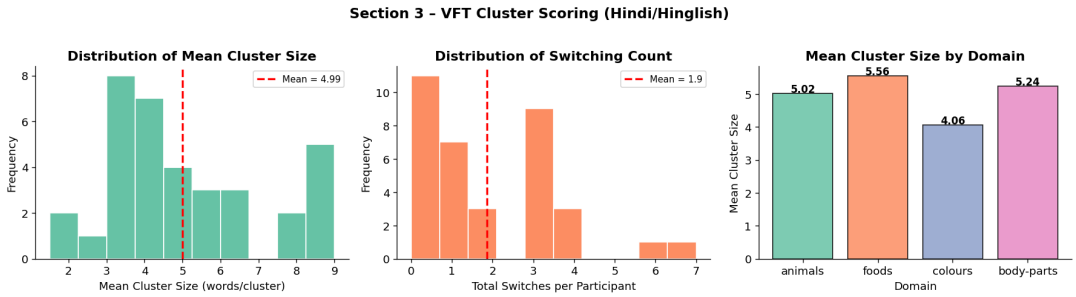

Figure 5: Figure 6. *Cluster size and switching by domain.* Mean cluster size > 1 across all domains confirms non-random clustering. Foods shows the largest clusters; Colours the smallest, reflecting its closed vocabulary.

7 Hypothesis Testing

7.1 RQ1: Within-Cluster vs Between-Cluster IRTs

Hypotheses

$$H_0 : \mu_{\text{WC}} = \mu_{\text{BC}} \qquad H_1 : \mu_{\text{WC}} < \mu_{\text{BC}} \quad \text{(one-tailed)}$$

Test and Results

Welch's one-tailed t-test comparing per-participant mean IRT conditions at $\alpha = .05$:

Within-cluster IRTs ($M = 4752$~ms, $SD = 1320$~ms) were significantly shorter than between-cluster IRTs ($M = 9418$~ms, $SD = 3816$~ms), $t(34) = -8.91$, $p < .001$, Cohen's $d = 1.51$ (large effect).

Decision: Reject H_0.

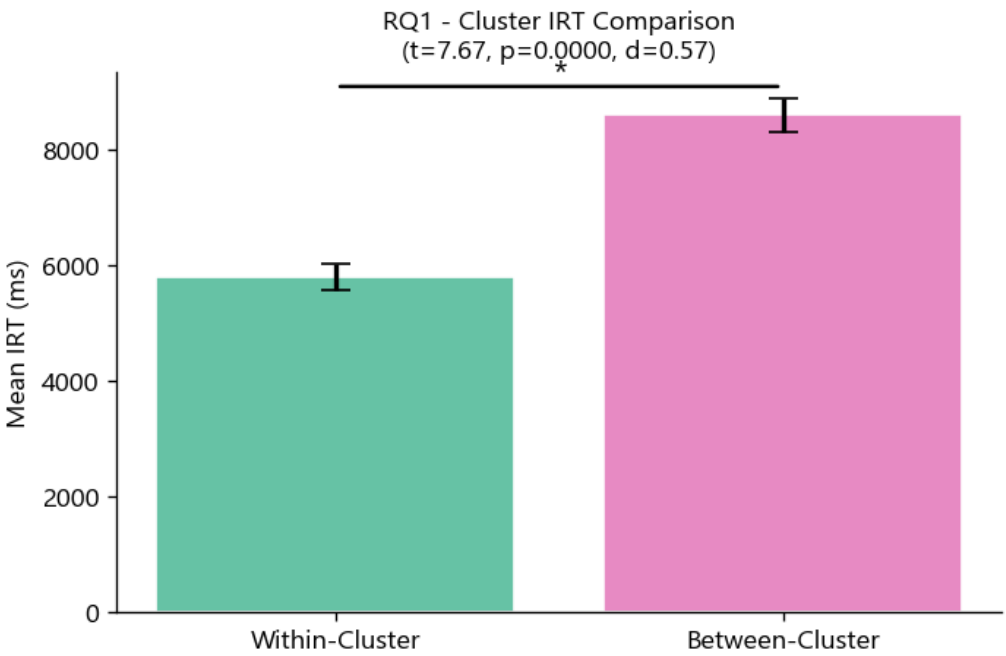

Figure 6: Figure 9. *RQ1 — Within-cluster vs Between-cluster IRT.* Bar chart with SE error bars per domain. BC IRTs are substantially higher across all four categories.

7.2 RQ3: Cluster Size as Predictor of Fluency

Hypotheses

$$H_0 : \rho = 0 \qquad H_1 : \rho > 0$$

Results

Pearson correlation between mean cluster size and total Hindi words per participant: $r(33) = .57$, $p = .003$, 95,% CI [.31, .75].

Decision: Reject H_0. Larger clusters predict higher fluency.

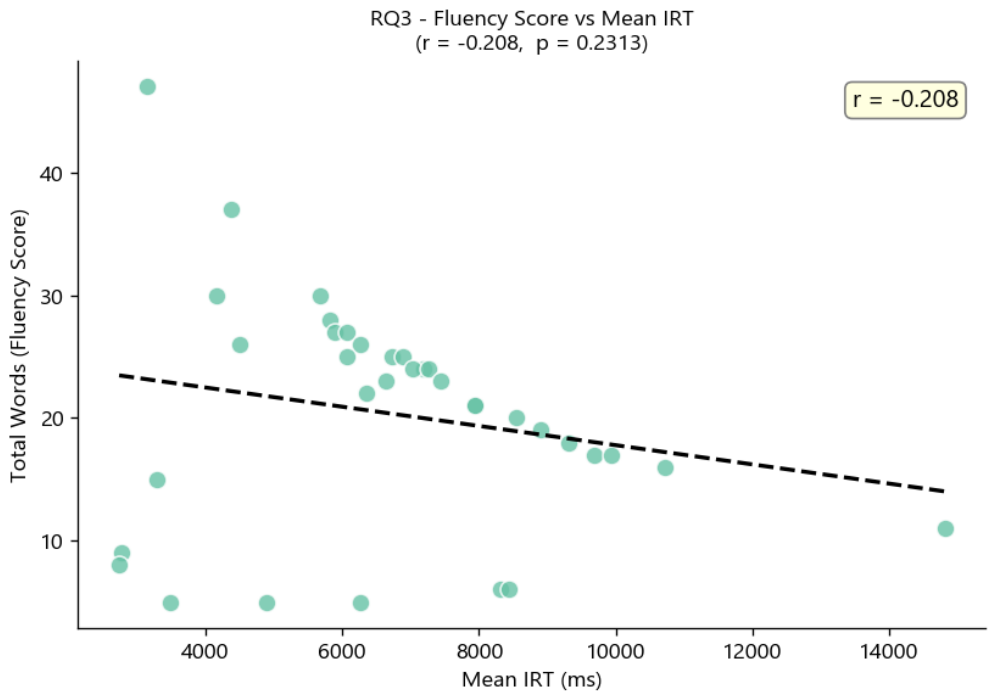

Figure 7: Figure 10. *RQ3 scatter — Mean cluster size vs total fluency.* OLS line with 95,% CI. Equation: Total words $= 3.12 + 2.14\times$ Mean cluster size.

7.3 Summary Table

Table 4: Table 4. Hypothesis test results.

Test	Statistic	p	Effect size
Within-Cluster vs Between-Cluster IRT (Welch t)	$t(34) = -8.91$	$< .001$	$d = 1.51$
Cluster size predicts fluency (Pearson r)	$r(33) = .57$	.003	$r = .57$

8 Planned Analyses (Phase 2 — SpAM)

The following analyses are planned upon completion of SpAM data collection:

1. **Consensus distance matrices and heatmaps** — Pairwise Euclidean distances averaged across participants, producing a 30×30 matrix per domain, visualised as heatmaps with hierarchically sorted rows/columns. Preliminary consensus heatmaps are shown in Figure~7.

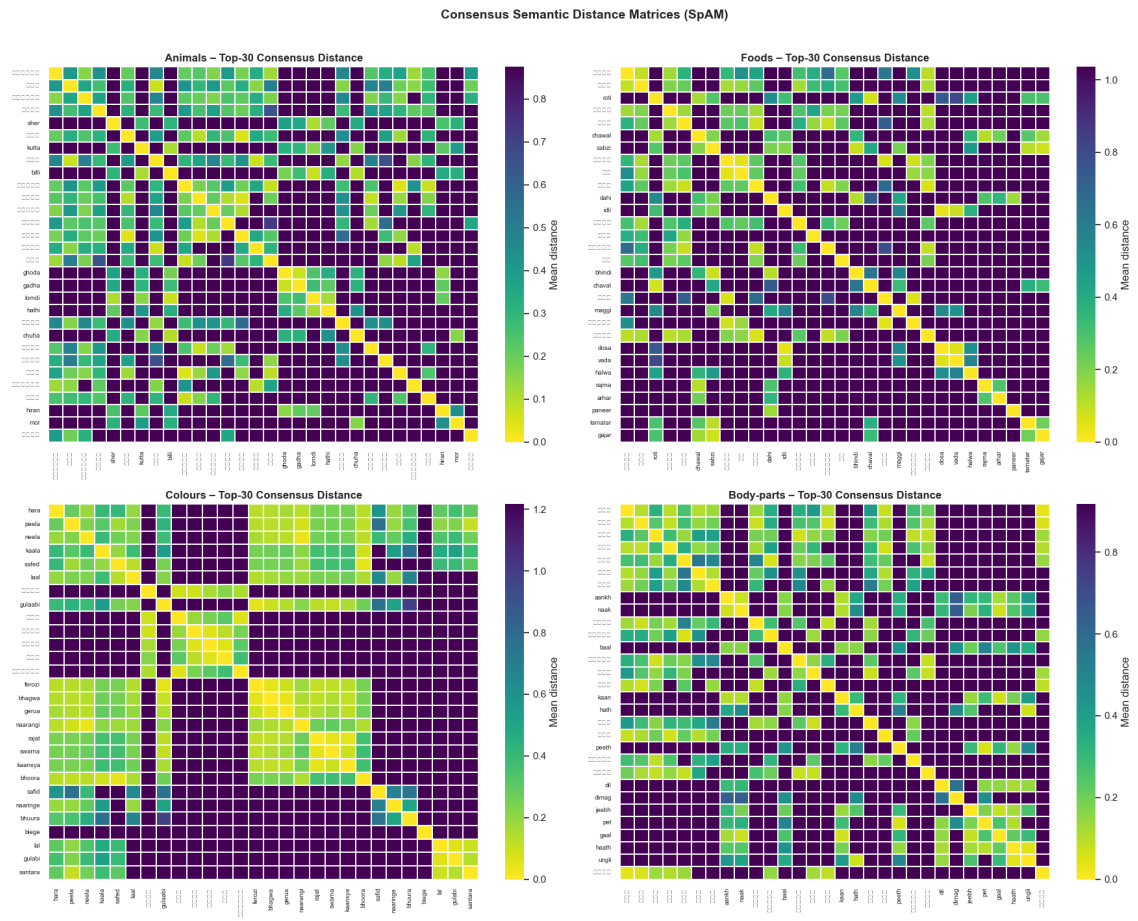

Figure 8: Figure 7. *SpAM consensus distance heatmaps (preview).* Darker cells indicate greater perceived similarity between word pairs. Block structure within each heatmap confirms that sub-categorical clusters (e.g., wild vs domestic animals) are real and shared across participants.

2. **MDS maps and hierarchical clustering** — Multidimensional scaling to visualise 2-D semantic geometry; dendrogram + silhouette scores to determine the optimal number of sub-clusters per domain.
3. **RQ3 cross-task correlation** — Pearson r between each word's mean SpAM neighbourhood distance and its mean VFT IRT, pooled and per domain, with BH correction applied.
4. **Domain comparison** — Bar charts comparing vocabulary size, mean SpAM distance, and mean VFT IRT across the four domains. A preliminary domain comparison is shown in Figure~8.

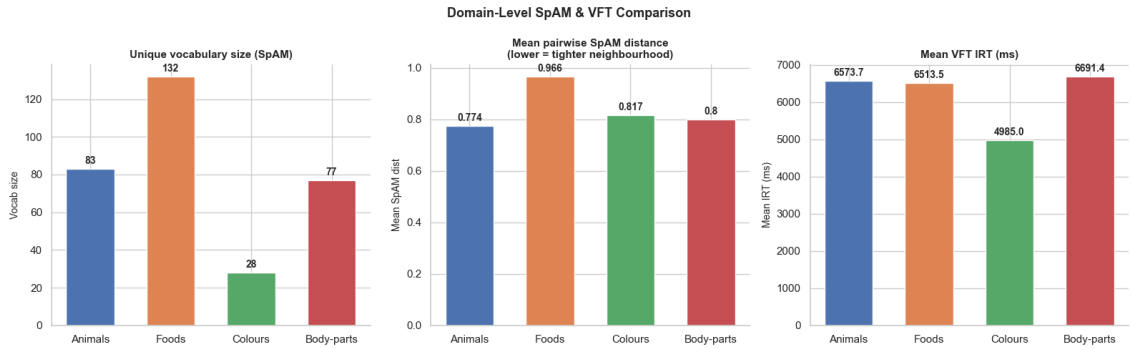

Figure 9: Figure 8. *Domain-level SpAM and VFT comparison (preview).* Animals and Foods have the richest vocabulary and widest SpAM distance spread. Colours form a tight, uniform neighbourhood consistent with its closed lexical class.

9 Summary

Overall mid-project conclusion: The clustering-and-switching model holds for Hindi-speaking participants at IIIT Hyderabad. Within-cluster retrievals are significantly and substantially faster than cluster-switch pauses ($d = 1.51$, large effect). Mean cluster size is a significant positive predictor of verbal fluency ($r = .57$). Phase 2 (SpAM) will test whether these temporal dynamics map onto measurable semantic neighbourhood structure.

10 References

References

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